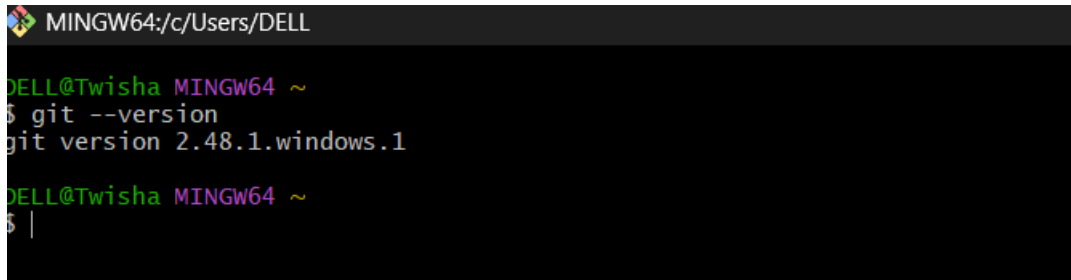
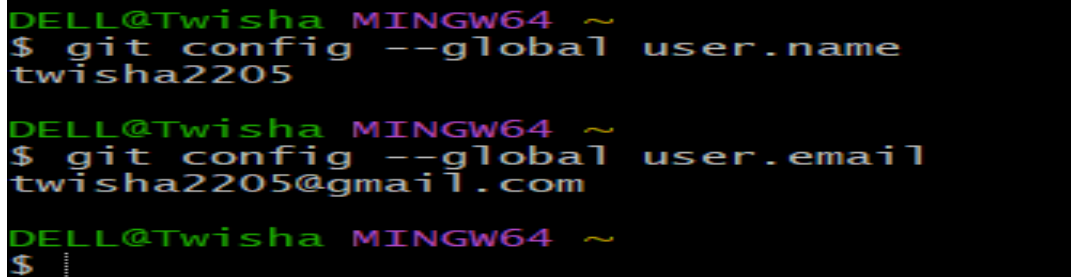


TASK-1

Git & GitHub Basics



```
MINGW64:/c/Users/DELL  
DELL@Twisha MINGW64 ~  
$ git --version  
git version 2.48.1.windows.1  
DELL@Twisha MINGW64 ~  
$ |
```



```
DELL@Twisha MINGW64 ~  
$ git config --global user.name  
twisha2205  
DELL@Twisha MINGW64 ~  
$ git config --global user.email  
twisha2205@gmail.com  
DELL@Twisha MINGW64 ~  
$
```

```
DELL@Twisha MINGW64 ~/desktop/internship
$ cd ecommerce-sales-analytics

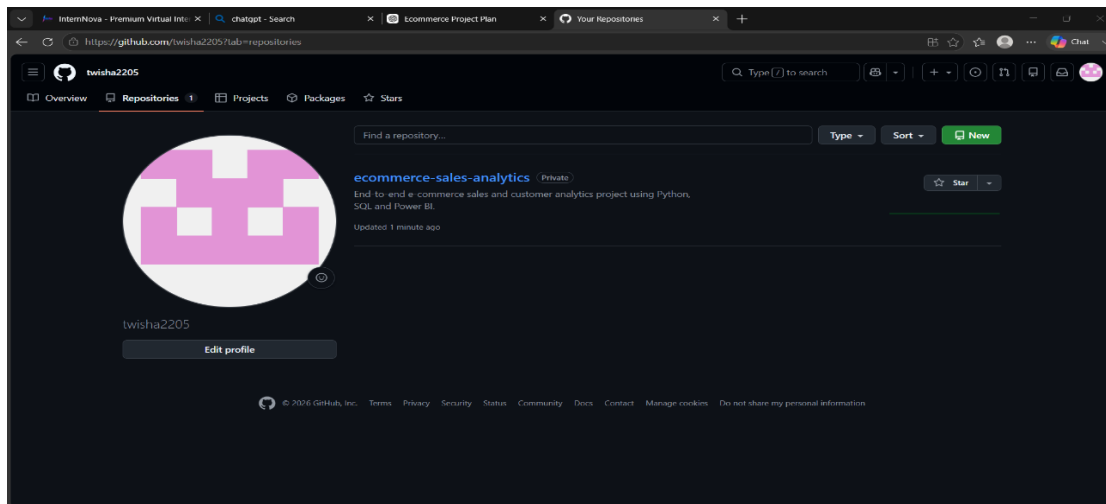
DELL@Twisha MINGW64 ~/desktop/internship/ecommerce-sales-analytics (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

DELL@Twisha MINGW64 ~/desktop/internship/ecommerce-sales-analytics (main)
$
```

```
DELL@Twisha MINGW64 ~/desktop/internship/ecommerce-sales-analytics (main)
$ git commit -m "Initial project setup"
[main f735b69] Initial project setup
1 file changed, 20 insertions(+), 2 deletions(-)

DELL@Twisha MINGW64 ~/desktop/internship/ecommerce-sales-analytics (main)
$
```



TASK-2

Git Workflow Practice

```
DELL@Twisha MINGW64 ~/desktop/internship/ecommerce-sales-analytics (main)
$ git add .

DELL@Twisha MINGW64 ~/desktop/internship/ecommerce-sales-analytics (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   .gitignore
```

[Staging project files using Git]

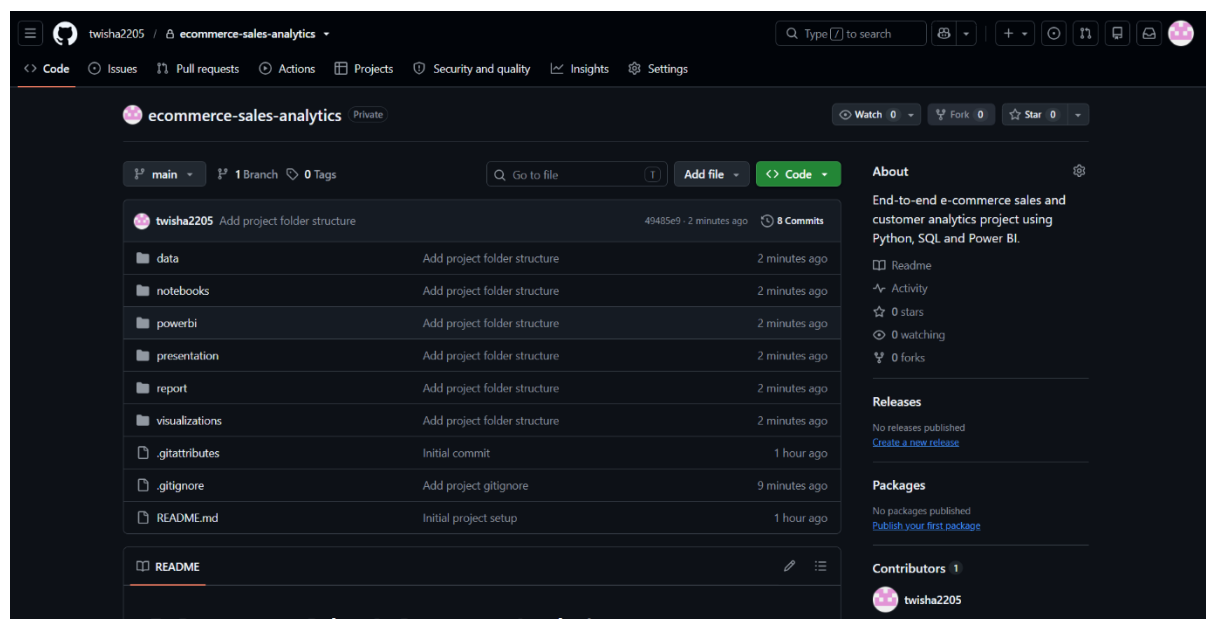
```
DELL@Twisha MINGW64 ~/desktop/internship/ecommerce-sales-analytics (main)
$ git commit -m "Add project gitignore"
[main e2c394d] Add project gitignore
1 file changed, 5 insertions(+)
create mode 100644 .gitignore
```

```
DELL@Twisha MINGW64 ~/desktop/internship/ecommerce-sales-analytics (main)
$ git commit -m "Add project folder structure"
[main 49485e9] Add project folder structure
7 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 data/cleaned/.gitkeep
create mode 100644 data/raw/.gitkeep
create mode 100644 notebooks/.gitkeep
create mode 100644 powerbi/.gitkeep
create mode 100644 presentation/.gitkeep
create mode 100644 report/.gitkeep
create mode 100644 visualizations/.gitkeep
```

[Creating a meaningful git commit]

```
DELL@Twisha MINGW64 ~/desktop/internship/ecommerce-sales-analytics (main)
$ git push origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 523 bytes | 174.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/twisha2205/ecommerce-sales-analytics.git
55a9354..49485e9  main -> main
```

[Pushing changes to GitHub]



[Github repository structure after push]

```
DELL@Twisha MINGW64 ~/desktop/internship/ecommerce-sales-analytics (main)
$ git pull
Already up to date.
```

[Git Pull command]

TASK-3

Dataset Selection & Data Preparation

Dataset Selection

Dataset Name: E-Commerce Sales Transactions Dataset

Source: Kaggle

Dataset File: ecommerce_sales_34500.csv

Description:

The dataset contains **34,500 e-commerce sales transactions** with **17 attributes** related to customers, products, orders, payments, delivery, returns, sales, and profitability. It includes information such as product category, price, discount, quantity, payment method, order date, region, total amount, shipping cost, profit margin, customer age, and gender.

Reason for Selection:

This dataset was selected because it represents real-world e-commerce transactions and provides sufficient information for analyzing **sales performance, customer behavior, product categories, discounts, returns, delivery time, and profitability**. It is suitable for data cleaning, exploratory data analysis, visualization, Power BI dashboard creation, and generating business insights.

Dataset Details

Attribute	Details
Source	Kaggle
Domain	E-Commerce
Records	34,500
Attributes	17

Attribute	Details
-----------	---------

File Format	CSV
-------------	-----

Analysis Tools Python, Pandas, Matplotlib, Seaborn, Power BI

Data Preparation:

Process:

1. Loaded the raw CSV dataset using Pandas.
2. Inspected the dataset dimensions, columns, and data types.
3. Checked for missing values and duplicate records.
4. Checked for invalid values in numerical columns.
5. Converted the order_date column into a proper datetime format.
6. Saved the prepared dataset as a separate cleaned CSV file.

Source code:

```
import pandas as pd
```

```
df = pd.read_csv("../data/raw/ecommerce_sales_34500.csv")
```

```
df.head()
```

```
#Total number of rows and columns in a dataset
```

```
print("Number of rows:", df.shape[0])
```

```
print("Number of columns:", df.shape[1])
```

```
#printing the columns and datatypes
```

```
df.info()
```

```
#finding missing values
```

```
print("Number of missing values in the dataset: ", df.isnull().sum())
```

```
#finding duplicate values
```

```
print("Number of duplicate rows:", df.duplicated().sum())
```

```
#checking for invalid/incorrect records
```

```
print("Negative prices:", (df["price"] < 0).sum())
```

```
print("Invalid discounts:", ((df["discount"] < 0) | (df["discount"] > 1)).sum())
```

```
print("Invalid quantities:", (df["quantity"] <= 0).sum())
```

```
print("Invalid delivery days:", (df["delivery_time_days"] <= 0).sum())
```

```
print("Invalid customer ages:", ((df["customer_age"] < 18) |  
(df["customer_age"] > 100)).sum())
```

```
print("Negative shipping costs:", (df["shipping_cost"] < 0).sum())
```

```
#converting date column
```

```
df["order_date"] = pd.to_datetime(df["order_date"], errors="coerce")
```

```
print(df["order_date"].dtype)
```

```
print(df["order_date"].head())
```

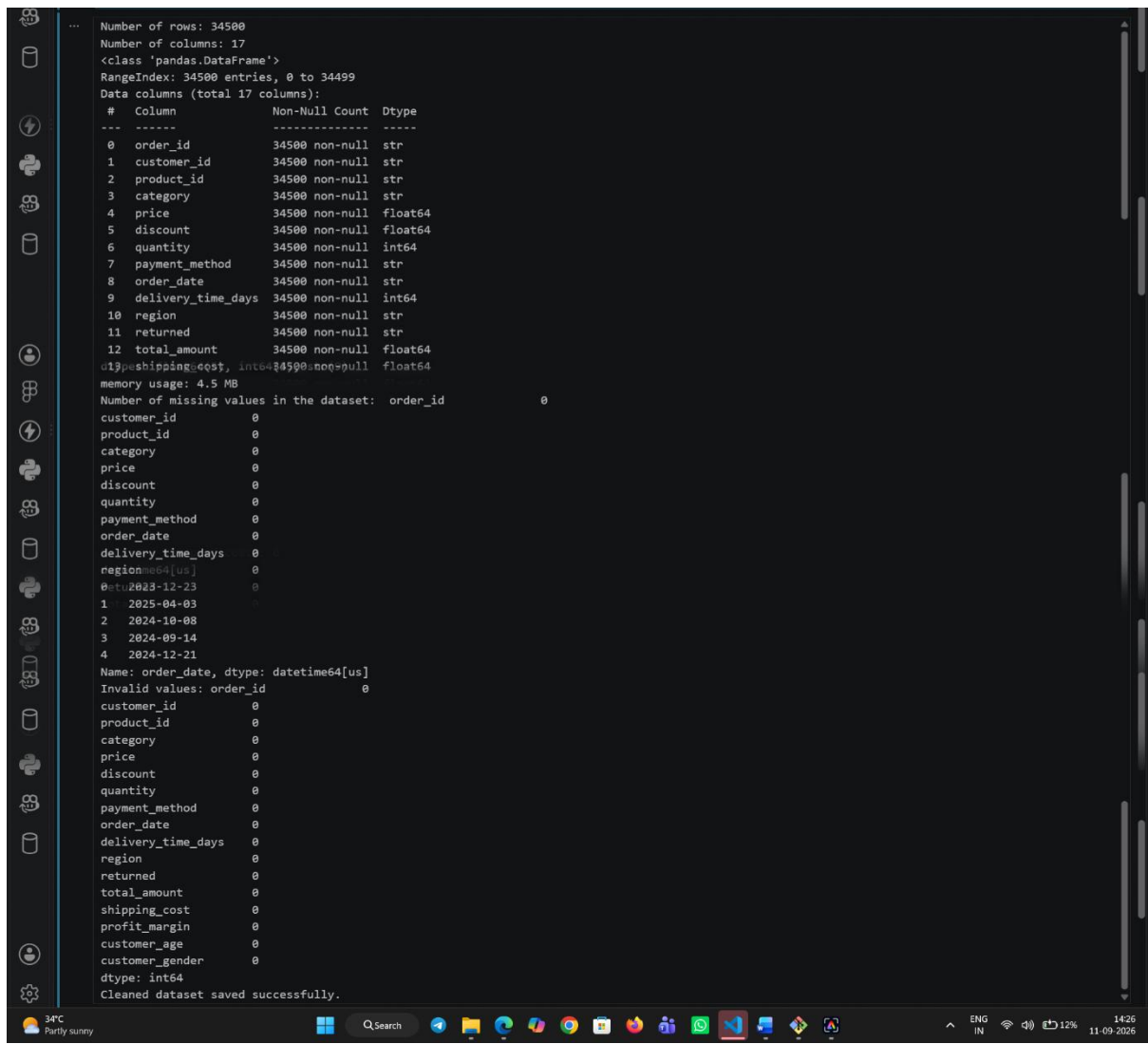
```
#checking missing value for order_date column
```

```
print("Invalid values:", df.isnull().sum())
```

#saving the cleaned dataset

```
df.to_csv("../data/cleaned/ecommerce_sales_cleaned.csv", index=False)
```

```
print("Cleaned dataset saved successfully.")
```



The screenshot shows a Jupyter Notebook interface with a terminal window displaying the following output:

```
Number of rows: 34500
Number of columns: 17
<class 'pandas.DataFrame'>
RangeIndex: 34500 entries, 0 to 34499
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   order_id               34500 non-null  str
1   customer_id            34500 non-null  str
2   product_id             34500 non-null  str
3   category               34500 non-null  str
4   price                  34500 non-null  float64
5   discount               34500 non-null  float64
6   quantity               34500 non-null  int64
7   payment_method         34500 non-null  str
8   order_date             34500 non-null  str
9   delivery_time_days     34500 non-null  int64
10  region                 34500 non-null  str
11  returned               34500 non-null  str
12  total_amount           34500 non-null  float64
13  shipping_cost          34500 non-null  float64
memory usage: 4.5 MB
Number of missing values in the dataset: order_id      0
customer_id      0
product_id       0
category         0
price            0
discount         0
quantity         0
payment_method   0
order_date       0
delivery_time_days 0
regionme64[us]   0
0  2023-12-23     0
1  2025-04-03     0
2  2024-10-08     0
3  2024-09-14     0
4  2024-12-21     0
Name: order_date, dtype: datetime64[us]
Invalid values: order_id      0
customer_id      0
product_id       0
category         0
price            0
discount         0
quantity         0
payment_method   0
order_date       0
delivery_time_days 0
region           0
returned         0
total_amount     0
shipping_cost     0
profit_margin    0
customer_age     0
customer_gender  0
dtype: int64
Cleaned dataset saved successfully.
```

The terminal output provides a comprehensive overview of the dataset's structure and content. It includes the number of rows (34500) and columns (17), the data types for each column, and the count of non-null values. It also shows the memory usage (4.5 MB) and the number of missing values (0) for each column. The output concludes with a confirmation message: "Cleaned dataset saved successfully."

TASK-4

Exploratory Data Analysis (EDA)

EDA Findings

1. **Overall Sales:** The total sales generated from the dataset are **586,5293.05**, with an average order value of approximately **170.00**.
2. **Category Performance:** **Electronics** has the highest sales at approximately **3,319,206.50**, followed by **Home** at **1,077,681.52**. **Grocery** has the lowest sales at approximately **82,000.51**.
3. **Regional Performance:** Sales vary across different regions, indicating differences in customer demand and purchasing patterns across regions.
4. **Profitability:** The overall average profit margin is approximately **28.12**. **Electronics** has the highest average profit margin at **55.72**, while **Grocery** has the lowest at approximately **-2.26**.
5. **Sales Trend:** Monthly sales fluctuate throughout the period. Sales reach their highest point around **December 2024**, at approximately **278,000**, while the final month shows a sharp decline to around **92,000**.
6. **Returns:** The overall return rate is approximately **5.52%**. This indicates that most orders were not returned, although return behavior may vary across categories and regions.
7. **Customer Analysis:** The dataset contains customers from different age groups and genders. Customer characteristics can be further analyzed to understand purchasing patterns and identify important customer segments.
8. **Correlation:** The correlation analysis shows relationships between numerical variables such as **price, quantity, discount, total amount**,

shipping cost, and profit margin. These relationships help identify factors that influence sales and profitability.

9. **Outliers:** The boxplot shows several outliers in **profit margin**, including some very high values. These were retained because they may represent genuine transactions rather than data errors.
10. **Data Quality:** The dataset contains **34,500 records and 17 columns**. No missing values or duplicate records were found, and the data was checked for invalid numerical values. The `order_date` column was also converted into the proper date format.

File Edit Selection View Go Run

notebooks

data_preparation.ipynb • eda.ipynb

Sign In

Python 3.12.0

Explorer

notebooks

.gitkeep

data_preparation.ipynb

eda.ipynb

eda.ipynb

Load and inspect the data

Generate Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline Python 3.12.0

print(return_by_region)

Customer analysis

print("Average Customer Age:", df["customer_age"].mean())

print("Minimum Customer Age:", df["customer_age"].min())

print("Maximum Customer Age:", df["customer_age"].max())

gender_sales = df.groupby("customer_gender")["total_amount"].sum().sort_values(ascending=False)

print("Sales by Gender:")

print(gender_sales)

[30] ✓ 0.7s Python

Dataset Shape: (34500, 17)

	price	discount	quantity	order_date	delivery_time_days	total_amount	shipping_cost	profit
count	34500.000000	34500.000000	34500.000000	34500	34500.000000	34500.000000	34500.000000	34500
mean	119.391632	0.049291	1.490725	2024-09-13 13:39:17.843478	4.814203	170.008494	6.152120	117.807373
min	1.010000	0.000000	1.000000	2023-09-12 20:00:00.000000	3.000000	0.820000	0.000000	0.820000
mean	119.391632	0.049291	1.490725	2024-09-13 13:39:17.843478	4.814203	170.008494	6.152120	117.807373
min	1.010000	0.000000	1.000000	2023-09-12 00:00:00.000000	3.000000	0.820000	0.000000	0.820000
25%	16.690000	0.000000	1.000000	2024-03-14 00:00:00.000000	4.000000	19.710000	4.420000	15.290000
50%	45.660000	0.000000	1.000000	2024-09-16 00:00:00.000000	5.000000	56.820000	6.090000	50.730000
75%	130.950000	0.100000	2.000000	2025-03-17 00:00:00.000000	6.000000	168.530000	7.830000	160.700000
max	2930.470000	0.300000	5.000000	2025-09-11 00:00:00.000000	13.000000	12931.800000	15.650000	12916.150000
std	195.620477	0.069894	0.932270	NaN	1.242141	357.503014	2.389539	193.218884

Categories: <StringArray>

['Home', 'Grocery', 'Electronics', 'Beauty', 'Fashion', 'Toys', 'Sports']

Length: 7, dtype: str

Regions: <StringArray>

['West', 'South', 'North', 'East', 'Central']

Length: 5, dtype: str

Payment Methods: <StringArray>

['Credit Card', 'UPI', 'COD', 'PayPal', 'Debit Card', 'Wallet']

Length: 6, dtype: str

Return Status: <StringArray>

['No', 'Yes']

Length: 2, dtype: str

Missing values: 0

Duplicate rows: 0

Negative prices: 0

Invalid discounts: 0

Invalid quantities: 0

Invalid delivery days: 0

Invalid customer ages: 0

Correlation Matrix of E-Commerce Variables

	price	discount	quantity	delivery_time_days	total_amount	shipping_cost	profit_margin
price	1.00	0.01	0.01	0.01	0.80	0.65	0.68
discount	0.01	1.00	-0.01	0.00	-0.03	-0.05	-0.05
quantity	0.01	-0.01	1.00	0.01	0.30	0.28	0.39
delivery_time_days	0.01	0.00	0.01	1.00	0.01	0.01	0.01
total_amount	0.80	-0.03	0.30	0.01	1.00	0.60	0.89
shipping_cost	0.65	-0.05	0.28	0.01	0.60	1.00	0.62
profit_margin	0.68	-0.05	0.39	0.01	0.89	0.62	1.00

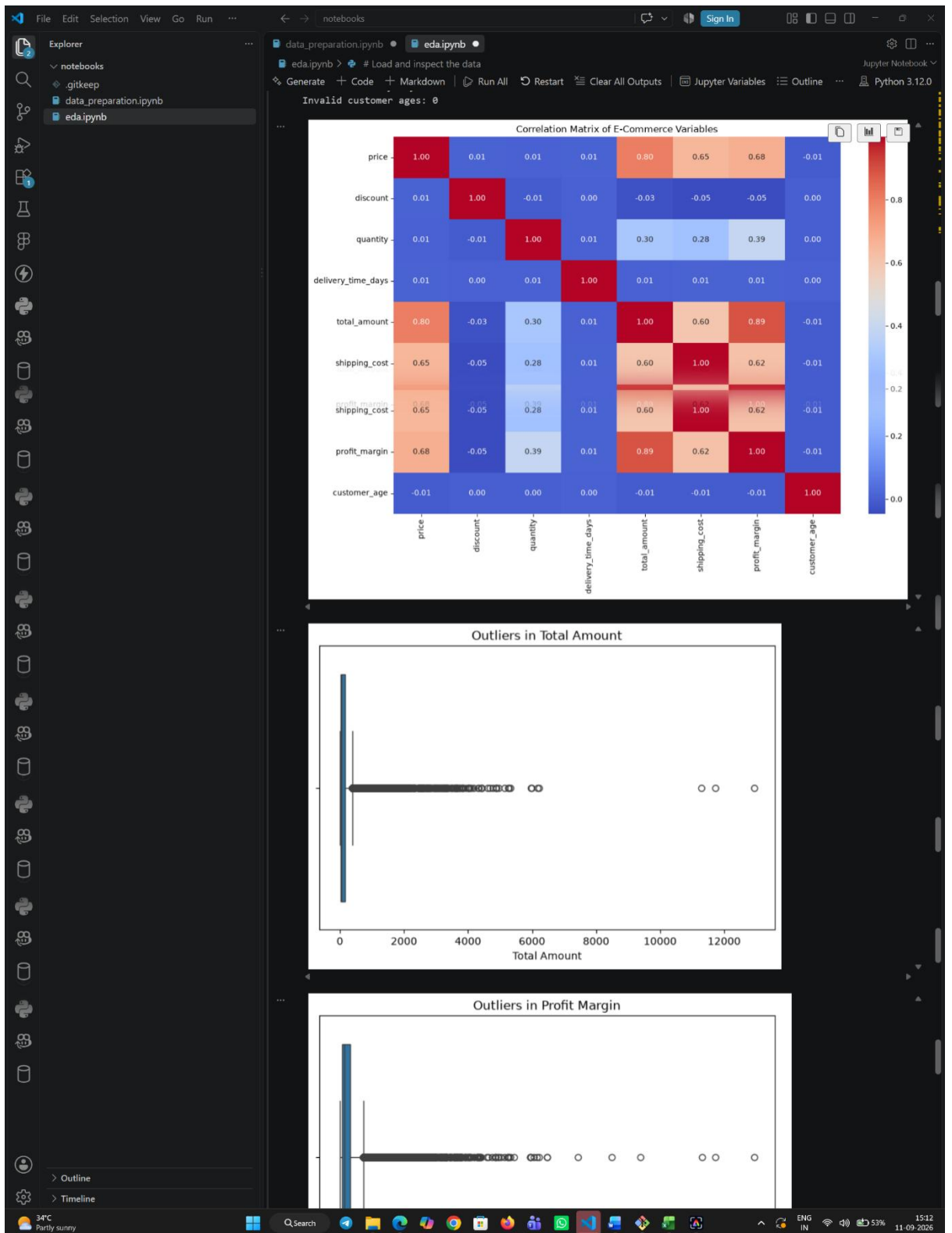
Outline

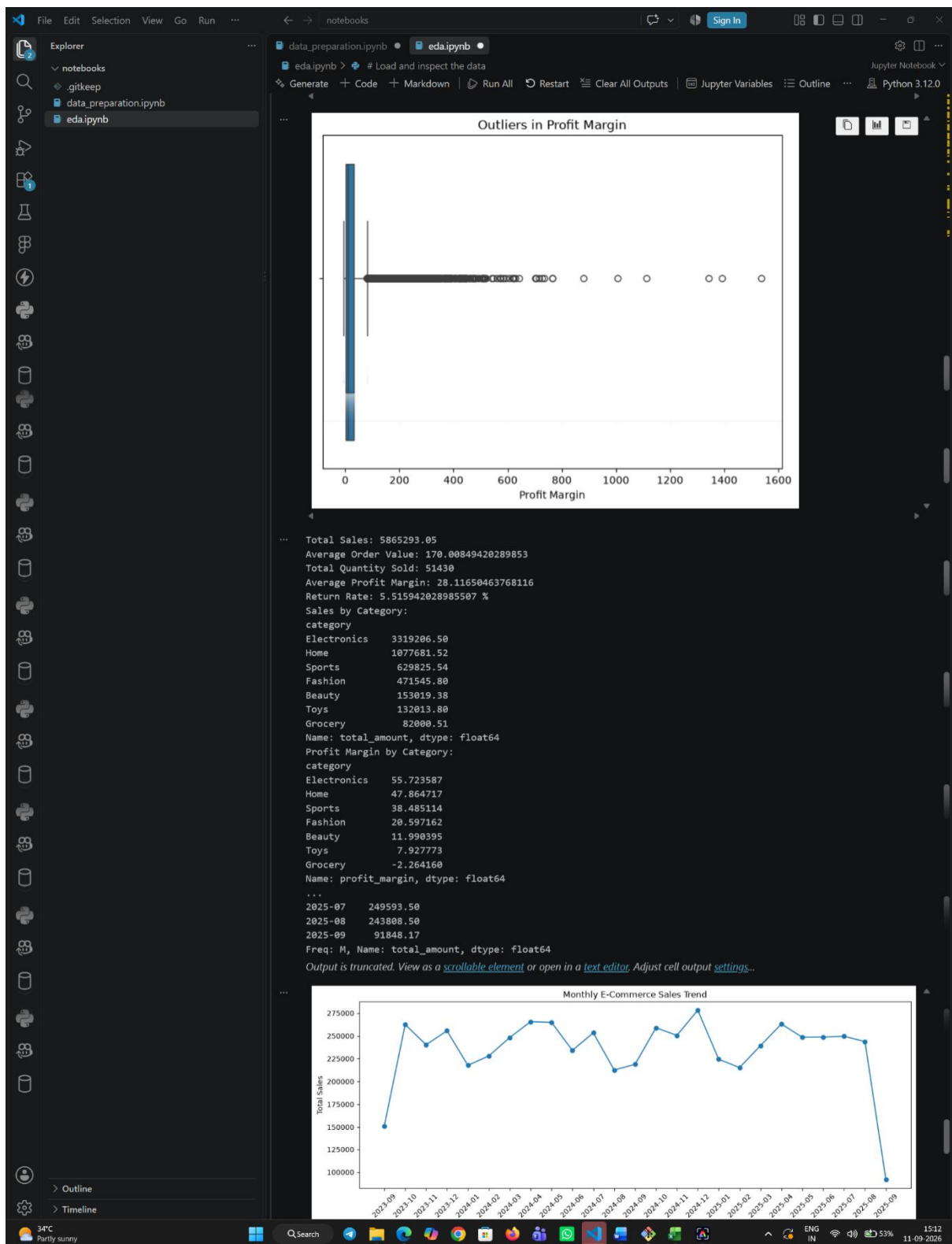
Timeline

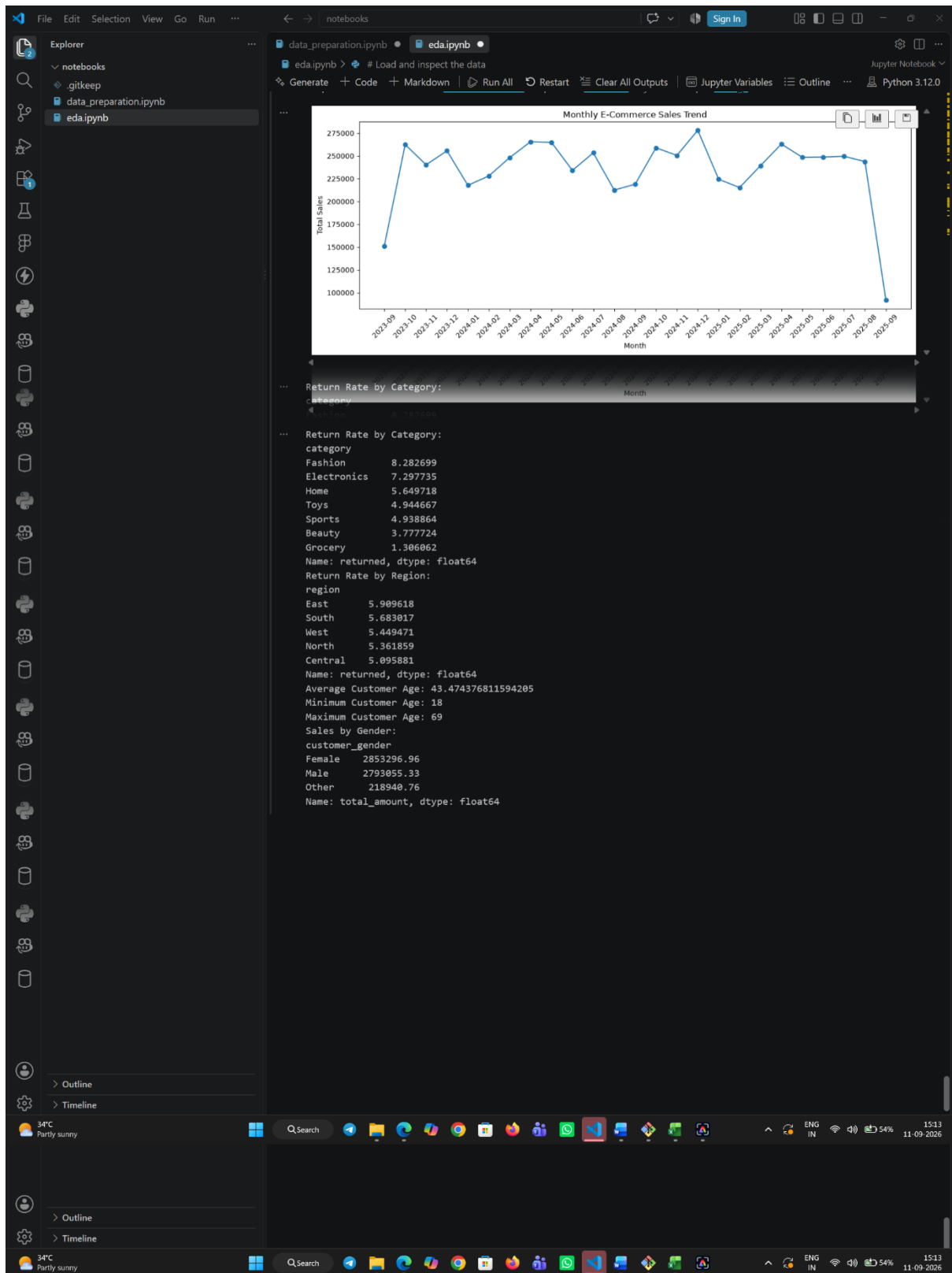
34°C Partly sunny

Search

ENG IN 52% 15:11 11-09-2026

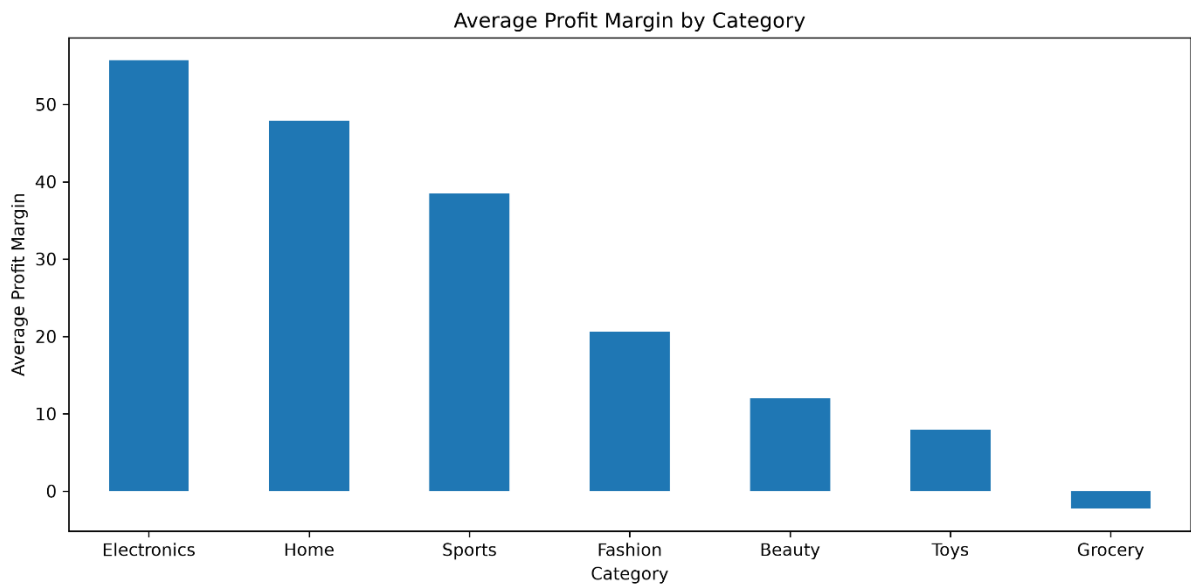


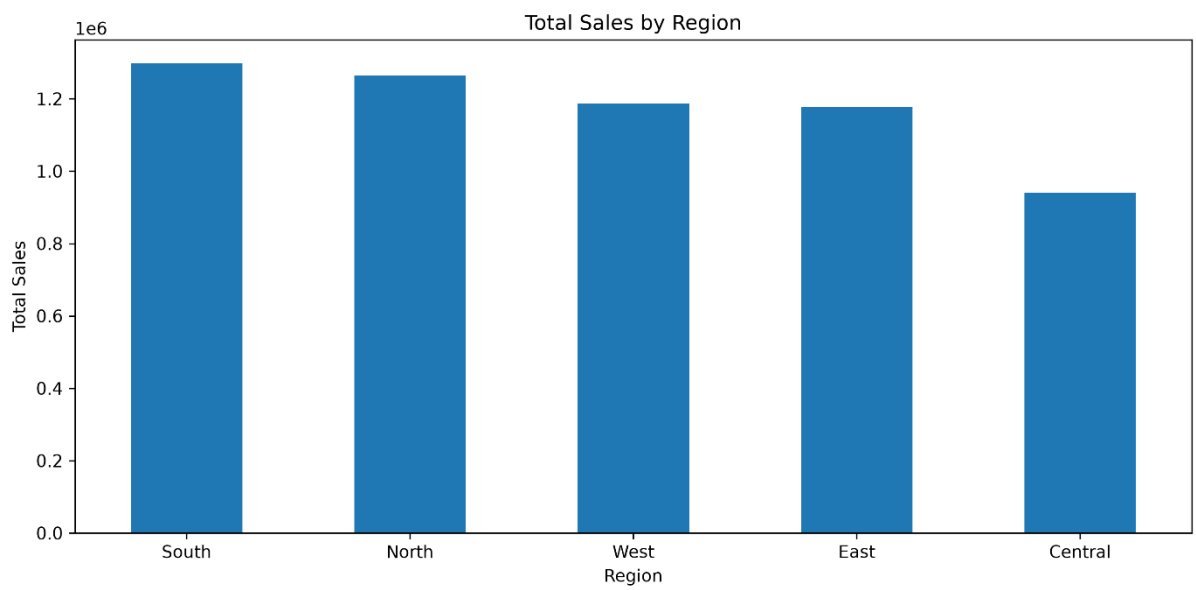
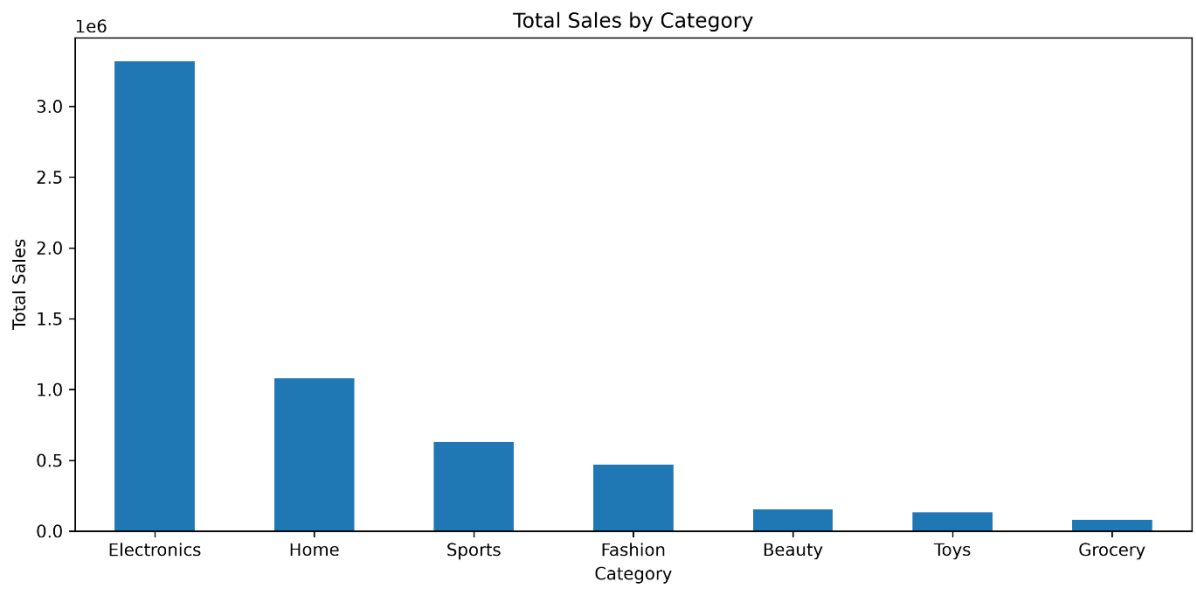
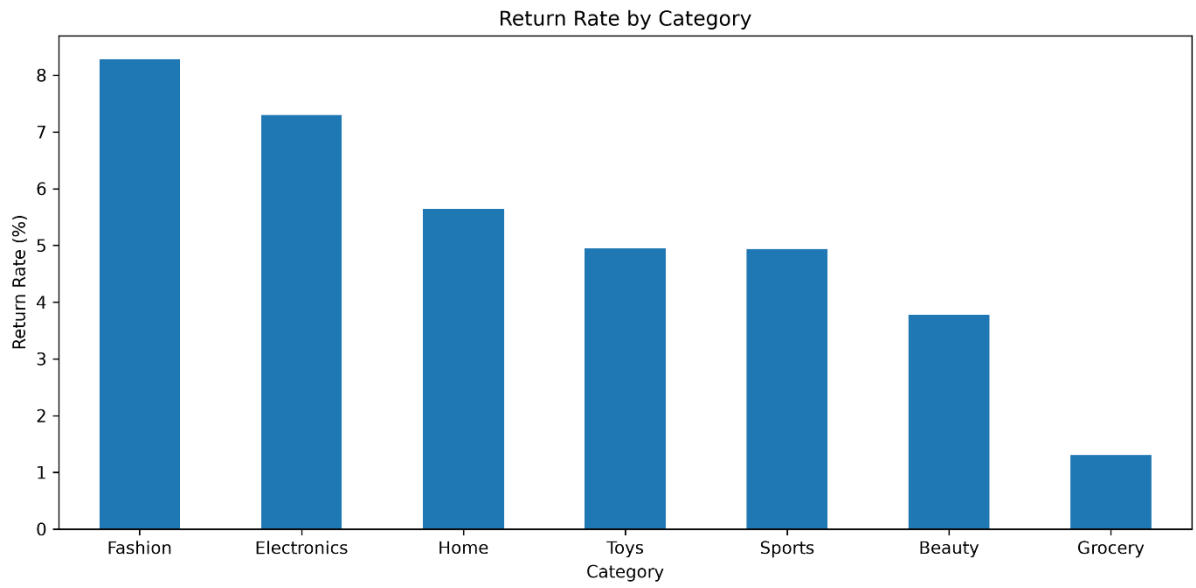




TASK-5

Data Visualization





TASK-6

Power BI Dashboard



Task-7

Business Insights & Recommendations

Insights:

Insight 1 — Electronics is the strongest sales category

Electronics generated approximately **3.32M in sales**, making it the highest-performing category in the dataset.

Insight 2 — Grocery has a profitability problem

Grocery generated approximately **82K in sales**, but its average profit margin was around **-2.26**, indicating that this category may be operating at a loss.

Insight 3 — Electronics also has strong profitability

Electronics recorded an average profit margin of approximately **55.72**, showing that it is not only a high-sales category but also highly profitable.

Insight 4 — Overall customer returns are relatively limited

The overall return rate was approximately **5.52%**, meaning most orders were not returned. However, category-level differences should still be monitored.

Insight 5 — Sales show noticeable monthly variation

Sales fluctuated considerably throughout the period, with sales reaching approximately **278K in December 2024**, indicating a strong seasonal/peak-period effect.

Insight 6 — Profitability varies significantly across categories

The difference between high-margin categories such as Electronics and low/negative-margin categories such as Grocery indicates that category-level profitability should be considered alongside sales volume.

Recommendations:

Recommendation 1 — Focus on high-performing categories

The business should continue investing in Electronics, including inventory availability, promotions and marketing, because it combines high sales with a strong profit margin.

Recommendation 2 — Review Grocery pricing and costs

The -2.26 average profit margin for Grocery requires investigation. The company should review pricing, supplier costs, discounts and shipping expenses to improve profitability.

Recommendation 3 — Prepare for seasonal demand

Since sales peaked around December 2024, the business should increase inventory and marketing efforts before high-demand periods and use historical sales trends for demand planning.

Recommendation 4 — Monitor product returns

Although the overall return rate is around 5.52%, management should identify categories or products with unusually high return rates and investigate the reasons behind them.

Recommendation 5 — Use category-level performance for decision-making

Instead of evaluating the business only on total sales, management should track sales, profit margin and return rate together to identify categories that generate sustainable revenue.